Why Per-Layer Residual Gating Fails to Fix Multi-Turn Attention Degradation: A Mechanistic Analysis

Anonymous ACL submission

Abstract

Multi-turn dialogue with Large Language Models (LLMs) suffers from progressive attention decay toward system prompt goal tokens. On Qwen-2.5-7B-Instruct, the Goal Accessibility Ratio (GAR) declines by 70.8% between early and mid-conversation turns, yet linear probes recover goal content from the residual stream with AUC = 0.98. We introduce ThreadGate, a per-layer content-and-turn-conditioned MLP gate ($x_{l+1} = x_l + g_l \cdot f_l(x_l)$) trained jointly with next-token prediction (NTP) and an auxiliary GAR loss. A five-way comparison reveals that training succeeds (teacher-forced GAR = 0.50) but autoregressive GAR drops to 0.015–0.025, a 20–33× gap; gating contributes zero measurable benefit and a PCGrad-based exclusion experiment rules out gradient conflict. The failure traces to exposure bias: attention patterns optimized under teacher-forced prefixes do not transfer to model-generated text. Inference-time steering ($\beta = 5$) recovers GAR from 0.064 to 0.727, providing constructive evidence that attention misallocation—not representation degradation—is a proximal cause consistent with our evidence. Scheduled sampling compresses the gap to 1.57× with no PPL cost (single seed), and cross-model evaluation on Llama-3.1-8B confirms the gap (27× by median). These findings characterize how exposure bias selectively destroys auxiliary objective generalization under the tested conditions (Qwen-2.5-7B and Llama-3.1-8B, UltraChat, GAR-family losses).

1 Introduction

Multi-turn dialogue is the dominant mode of interaction with large language models (LLMs), and system prompts that specify task goals form the foundation of instruction following. As conversations grow longer, however, models progressively lose track of these goal specifications. Dongre et al. (2026) quantified this degradation across 12 models: the Goal Accessibility Ratio (GAR), which measures how much attention the model allocates to goal tokens in the system prompt, declines by 27–48% between early and mid-conversation turns. We replicate and extend this finding on Qwen-2.5-7B-Instruct, observing a 70.8% relative decline in GAR from 0.0418 at turn 1 to 0.0122 at mid-conversation. We attribute the steeper decline partly to our use of last-token GAR, which captures summary-position attention more sensitively than the all-token average of Dongre et al. (2026), and partly to Qwen-2.5-7B's stronger instruction-following fidelity in early turns. This attention decay represents a systematic failure of multi-turn models to maintain task coherence, and it grows worse with each additional turn.

A striking aspect of this degradation is what it does *not* affect. Dongre et al. (2026) showed that goal-relevant information persists in the residual stream throughout the conversation: linear discriminant analysis probes recover goal content with AUC = 0.98 even at late turns where attention to goal tokens has collapsed. The information is present in the hidden states, but the attention mechanism fails to route it toward the output computation. This dissociation between information availability and information utilization suggests a natural intervention point.

If the residual stream already carries goal information, a learnable gate on the residual pathway could amplify or attenuate layer contributions to recover goal attention. This reasoning motivates ThreadGate, which modulates each residual update with a content-and-turn-conditioned MLP gate: $x_{l+1} = x_l + g_l \cdot f_l(x_l)$, where g_l is produced by a two-layer network conditioned on the hidden state and turn features, inspired by the residual scaling idea of ReZero (Bachlechner et al., 2021).

ThreadGate combines per-layer MLP gate networks with a joint training objective: next-token prediction (NTP), an auxiliary GAR loss ($\tau = 0.5$), and a gate diversity loss (§3). The final formulation

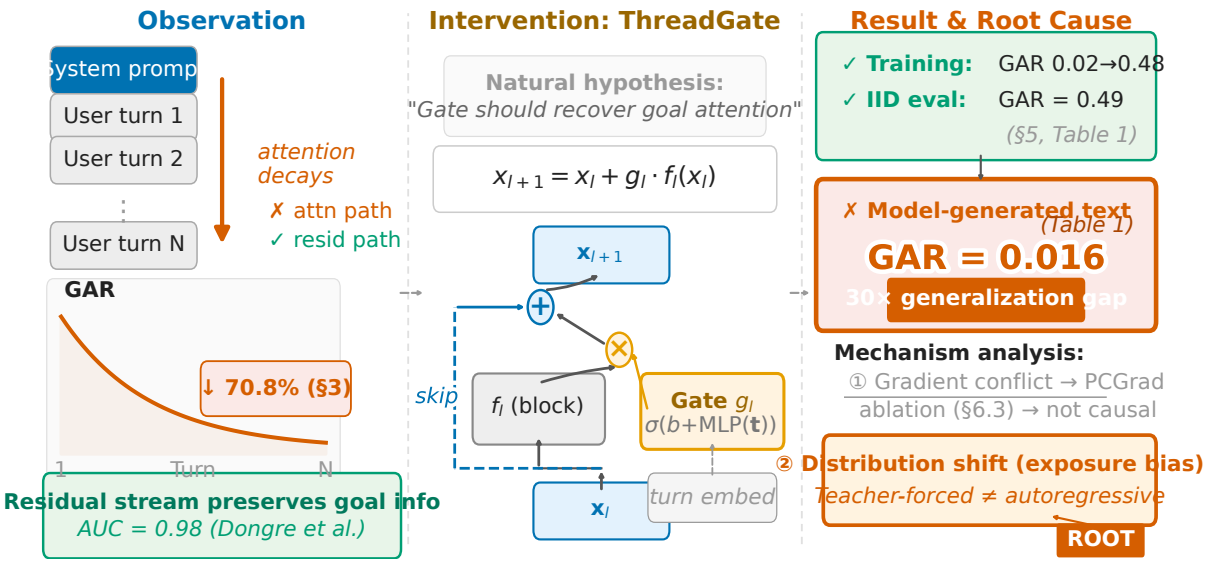


Figure 1: Overview of ThreadGate and its failure mode. (Left) ThreadGate adds per-layer content-and-turn-conditioned MLP gates to the residual stream, trained jointly with an auxiliary GAR loss that targets goal attention. (Right) Under teacher-forced evaluation, GAR-ours reaches 0.50 across all fine-tuned variants; under autoregressive generation, it drops to 0.015–0.025, a 20–33× gap. The five-way comparison (ThreadGate, +PCGrad, Uniform Gate, LoRA-only, Base) shows that no intervention provides measurable benefit over LoRA-only fine-tuning.

emerged from seven iterations spanning four loss families; only MSE target with gate diversity converged stably. To diagnose the failure, we construct a five-way controlled comparison (ThreadGate, +PCGrad, Uniform Gate, LoRA-only, Base) where each pair isolates one factor (§4). PCGrad (Yu et al., 2020) serves a dual role: training strategy and exclusion tool that rules out gradient direction conflict.

Training succeeds: teacher-forced GAR-ours reaches 0.50 across all fine-tuned variants. The five-way comparison reveals that the failure is specific to autoregressive generation, where GAR-ours drops to 0.015–0.025, a gap exceeding an order of magnitude that persists across all training strategies. All four fine-tuned variants achieve indistinguishable performance (0.75 pp spread on GAR-ours, identical PPL at 2.78); the gating contrast confirms that gates contribute zero measurable benefit. The PCGrad exclusion experiment eliminates gradient direction conflict without altering any outcome, establishing that conflict is neither necessary nor sufficient for the failure. These contrasts converge on exposure bias (Bengio et al., 2015), where attention patterns optimized under teacher-forced prefixes do not transfer to model-generated text (Figure 1). An inference-time steering experiment provides constructive evidence that attention misallocation is a proximal cause: additive attention bias recovers autoregressive GAR from 0.064 to 0.727, bypassing the exposure bias barrier (§4.3). A preliminary scheduled sampling experiment confirms that the gap is also mitigable at training time: GAR-ours increases from 0.016 to 0.318 (20×), compressing the ∼31× gap to 1.57× with no PPL cost (single seed; §6).

Our contributions: (1) a systematic diagnostic framework for analyzing auxiliary objective failures in multi-turn dialogue, comprising a five-way controlled comparison that isolates four factors (turn conditioning, gating mechanism, gradient direction conflict, and fine-tuning), PCGrad-based exclusion via gradient surgery (Yu et al., 2020), and per-layer gradient conflict quantification (§3, §4, §5); (2) using this framework, to our knowledge the first empirical characterization of how exposure bias selectively destroys auxiliary attention loss generalization (20–33× gap) while leaving NTP robust, with evidence that token-level self-recovery (He et al., 2019) does not extend to auxiliary objectives, a preliminary demonstration that scheduled sampling compresses this gap from ∼31× to 1.57× (single seed), and cross-model confirmation on Llama-3.1-8B (27× gap); (3) an inference-time steering experiment: additive attention bias at $\beta = 5$ recovers autoregressive GAR from 0.064 to 0.727, providing constructive evidence that attention misallocation—not represen-

tation degradation—is a proximal cause under the tested conditions (Qwen-2.5-7B and Llama-3.1-8B, UltraChat, GAR-family losses; §4.3); (4) a principled negative baseline: seven loss configurations across four families (Table 2; Appendix B), of which only MSE converges, and a five-way comparison ruling out four alternative training strategies (§3, §4).

2 Related Work

Multi-turn attention degradation. Recent work has established that LLMs systematically degrade across conversation turns. Laban et al. (2025) showed that model performance drops on tasks placed in later turns, even when the tasks are identical to earlier ones. Liu et al. (2026) traced a related failure to intent mismatch, where models progressively shift from the user's original intent as dialogue extends. MT-Bench-101 (Bai et al., 2024) provides fine-grained multi-turn evaluation across 1388 turn-level instances, revealing that capability varies substantially across task categories. Dongre et al. (2026) established the quantitative foundation for this line of work: they defined the Goal Accessibility Ratio (GAR) and showed that attention to system-prompt goal tokens decays by 27–48% across turns in 12 models, while residual stream probes retain goal information with AUC = 0.98. We observe a steeper 70.8% decline on Qwen-2.5-7B-Instruct (§4). This dissociation between attention collapse and residual persistence motivates our gating intervention. Their work characterizes the phenomenon; ours attempts the first systematic cure and, in doing so, uncovers why lightweight interventions fail to transfer from training to inference.

Instruction stability. Li et al. (2024) independently quantified instruction instability in multi-turn dialogs and proposed split-softmax—separating instruction from conversation tokens during attention computation—as an inference-time remedy. Split-softmax restructures the attention mechanism itself and requires no retraining, but cannot address exposure bias in auxiliary losses that depend on training-time optimization. Our work adds systematic diagnosis (five-way controlled comparison + PCGrad exclusion), exposure bias attribution, and training-time mitigation via scheduled sampling; our β-steering complements split-softmax as an alternative inference-time approach (§6).

Residual gating and adaptive depth. ReZero (Bachlechner et al., 2021) showed that residual scaling shapes optimization dynamics. Subsequent work targets computational efficiency: LayerDrop (Fan et al., 2020), Mixture-of-Depths (Raposo et al., 2024), FlexiDepth (Luo et al., 2025), and CALM (Schuster et al., 2022). Gromov et al. (2025) showed that removing up to half a model's layers preserves most capability, paralleling our finding that LoRA-only matches all gating variants. ThreadGate applies residual gating to a different objective: recovering attention patterns in multi-turn dialogue.

Attention manipulation. StreamingLLM (Xiao et al., 2024) identified attention sinks and proposed anchor token retention; Hsieh et al. (2024) showed positional bias against middle-positioned information; Qiu et al. (2025) replaced softmax with a learned gate for sparse attention; Rhea (Hong et al., 2025) introduces role-aware episodic attention for multi-turn dialogue. These approaches intervene inside the attention mechanism, while ThreadGate operates on the residual stream. Our results suggest that neither pathway alone overcomes the distributional gap between teacher-forced training and autoregressive inference.

Multi-task gradient conflict. GradNorm (Chen et al., 2018), PCGrad (Yu et al., 2020), CAGrad (Liu et al., 2021), and Nash-MTL (Navon et al., 2022) treat gradient conflict between competing losses as a root cause to be resolved through better optimization. We repurpose PCGrad as a diagnostic tool: eliminating directional conflict between NTP and GAR gradients does not improve GAR convergence. Gradient conflict is a symptom rather than the cause.

Exposure bias. Teacher forcing creates a distributional mismatch between training and inference that compounds over sequence length (Wiseman and Rush, 2016; Arora et al., 2022). Scheduled Sampling (Bengio et al., 2015) introduced curriculum-based mixing of ground-truth and model-generated tokens. MIXER (Ranzato et al., 2016) combined token-level and sequence-level objectives, and Professor Forcing (Lamb et al., 2016) used adversarial regularization to align hidden state distributions under both regimes. He et al. (2019) showed that autoregressive models exhibit self-recovery from exposure bias at the token prediction level. SEARNN (Leblond et al., 2018) pro-

posed locally recurrent search-based training to avoid the mismatch entirely. This literature focuses on generation quality degradation in primary objectives. No prior work examines how exposure bias affects auxiliary objectives that target attention distributions, nor how it manifests in multi-turn dialogue settings. Our work identifies a distinct failure mode: auxiliary attention losses that converge under teacher forcing fail completely under autoregressive generation, even when NTP remains robust, because token-level self-recovery does not extend to distributional properties of attention.

3 Method

3.1 Preliminaries

Goal Accessibility Ratio. We denote the task specification tokens in the system prompt as *goal tokens* $\mathcal{G}$. Dongre et al. (2026) defined the Goal Accessibility Ratio (GAR) for a model with L layers and H heads at turn τ as

$$\mathrm{GAR}(\tau) = \frac{1}{L \cdot H \cdot |\mathcal{R}_\tau|} \sum_{\ell=1}^{L} \sum_{h=1}^{H} \sum_{i \in \mathcal{R}_\tau} \sum_{j \in \mathcal{G}} A_{i,j}^{(\ell,h)}, \tag{1}$$

where $\mathcal{R}_\tau$ is the set of response tokens at turn τ and $A_{i,j}^{(\ell,h)}$ is the attention weight from token i to token j at layer ℓ, head h.

Two GAR variants. **GAR-Dongre** is the original definition (Eq. 1), averaging over all response tokens. **GAR-ours** restricts the measurement to the *last token* of each assistant turn:

$$\text{GAR-ours}(\tau) = \frac{1}{L \cdot H} \sum_{\ell=1}^{L} \sum_{h=1}^{H} \sum_{j \in \mathcal{G}} A_{t_\tau, j}^{(\ell,h)}, \tag{2}$$

where t_τ is the last token index at turn τ. Attention at this position reflects the information drawn upon for continuation; we use GAR-ours as the optimization target for $\mathcal{L}_{\text{GAR}}$ (§3.3).

3.2 ThreadGate Architecture

Inspired by the residual scaling idea of ReZero (Bachlechner et al., 2021), ThreadGate modulates the residual stream with a content-and-turn-conditioned gate at each transformer layer. The gated residual update at layer l is

$$x_{l+1} = x_l + g_l \cdot f_l(x_l), \tag{3}$$

where f_l denotes the sublayer computation (self-attention or FFN). The gate g_l is produced by a two-layer MLP:

$$g_l = \sigma\big(W_2\,\mathrm{ReLU}(W_1\,[\mathrm{proj}(x_l);\ \mathrm{emb}(t_d, t_i)] + b_1) + b_2\big), \tag{4}$$

where $\mathrm{proj}(\cdot)$ projects from the hidden dimension to 16 units and $\mathrm{emb}(t_d, t_i)$ is a 16-dimensional turn embedding encoding turn distance t_d (turns since system prompt) and turn index t_i (absolute position) via learned lookup tables. The concatenated 32-dimensional input passes through a 64-unit hidden layer to produce a scalar gate. Each of the 28 layers in Qwen-2.5-7B-Instruct receives its own gate network. Gates are initialized near 0.82 to preserve pretrained computation, with a $10\times$ learning rate multiplier. For the Uniform Gate ablation, $\mathrm{emb}(\cdot)$ is replaced by a learnable constant vector, preserving gate capacity while removing turn awareness.

3.3 Training Objective

The training objective combines three terms:

$$\mathcal{L} = \mathcal{L}_{\text{NTP}} + \lambda_{\text{GAR}}\,\mathcal{L}_{\text{GAR}} + \lambda_{\text{div}}\,\mathcal{L}_{\text{div}}. \tag{5}$$

$\mathcal{L}_{\text{NTP}}$ is the standard next-token prediction loss. The GAR loss targets a fixed attention level to goal tokens:

$$\mathcal{L}_{\text{GAR}} = \mathrm{MSE}(\mathrm{GAR}, \tau), \tag{6}$$

where $\tau = 0.5$. GAR is measured at the last token of each assistant turn (GAR-ours, §3.1). The gate diversity loss prevents collapse to a uniform value:

$$\mathcal{L}_{\text{div}} = \mathrm{ReLU}(\sigma_{\min} - \mathrm{std}(\mathbf{g})), \tag{7}$$

where $\mathbf{g} = [g_1, \ldots, g_L]$ and $\sigma_{\min}$ is a minimum standard deviation threshold. Both $\lambda_{\text{GAR}} = \lambda_{\text{div}} = 1.0$. We additionally tested three alternative loss families (cumulative, maximize, smooth saturation) across six configurations; only the MSE variant converged during training (Appendix B).

4 Experiments

4.1 Setup

We fine-tune Qwen-2.5-7B-Instruct on 8K multi-turn conversations from the UltraChat training split, a dataset of naturalistic dialogues across diverse topics. LoRA adapters (rank 16, α=32) are applied to query and value projection layers (`q_proj`, `v_proj`). The combined loss $\mathcal{L} = \mathcal{L}_{\text{NTP}} + \lambda_{\text{GAR}}\mathcal{L}_{\text{GAR}} + \lambda_{\text{div}}\mathcal{L}_{\text{div}}$ uses $\lambda_{\text{GAR}} = \lambda_{\text{div}} = 1.0$ with MSE target $\tau = 0.5$ (§3.3). We train for 3 epochs with AdamW (lr $= 2 \times 10^{-4}$, batch size 1,

gradient accumulation 8, seed 42; ThreadGate additionally with seed 123); each variant requires approximately 6 hours on a single NVIDIA RTX PRO 6000 Blackwell Server Edition GPU (102 GB). Full hyperparameters are provided in Appendix A.

We evaluate five configurations that form a controlled comparison isolating four factors. **ThreadGate**: full system with 28 gate networks (Eq. 4) and three-term loss. **+PCGrad**: ThreadGate with PCGrad (Yu et al., 2020) projecting conflicting gradients (isolates gradient conflict). **Uniform Gate**: identical gate architecture with turn embedding replaced by a still-learnable constant vector (content-conditioned but turn-unaware; isolates turn conditioning). **LoRA-only**: gates fixed at 1.0, trained with $\mathcal{L}_{\text{NTP}} + \mathcal{L}_{\text{GAR}}$ only (isolates gating mechanism). **Base**: pretrained model (isolates fine-tuning). All fine-tuned variants share training data, LoRA configuration, optimizer, and seed. We evaluate under *IID* (same-distribution UltraChat text) and *held-out* (50 non-overlapping conversations) teacher-forced protocols, reporting GAR-ours (last-token attention to goals, §3.1), GAR-Dongre (Dongre et al., 2026) (all-token average), and PPL.

4.2 Main Results

Table 1 presents the five-way comparison. The central finding is that all four fine-tuned variants converge to nearly identical performance. GAR-ours spans only 0.75 percentage points (pp) across ThreadGate (0.50), +PCGrad (0.50), Uniform Gate (0.51), and LoRA-only (0.50), and PPL is 2.78 for all four on IID evaluation. Neither per-layer gating (ThreadGate), elimination of gradient direction conflict (+PCGrad), nor content-only gating (Uniform Gate) provides any measurable benefit over LoRA-only fine-tuning.

The Base model confirms that LoRA fine-tuning is the operative variable: GAR-ours jumps from 0.01 to ∼0.50, while PPL improves from 4.23 to 2.78. Since all four variants share the same GAR loss ($\tau = 0.5$), convergence near the target is expected and does not demonstrate gating benefit. We cannot exclude LoRA capacity saturation at rank 16, though uniform convergence across four architecturally distinct variants makes this unlikely. GAR-Dongre remains uniformly low (0.01–0.02) across all standard-trained conditions, consistent with Dongre et al. (2026); a preliminary scheduled sampling experiment breaks this pattern (§6).

The teacher-forced GAR-ours of 0.50–0.51 stands in sharp contrast to autoregressive generation, where GAR-ours drops to 0.015–0.025 across all fine-tuned variants, a 20–33× gap. This gap is universal: it persists across all training strategies—gating, gradient conflict elimination, and LoRA-only fine-tuning alike. A second ThreadGate seed (123) confirms stability: TF GAR-ours 0.504 ± 0.001, AR GAR-ours 0.016 ± 0.001, gap ∼31× (range 30.5–32.1×), with identical PPL. We analyze this distribution shift in §5.

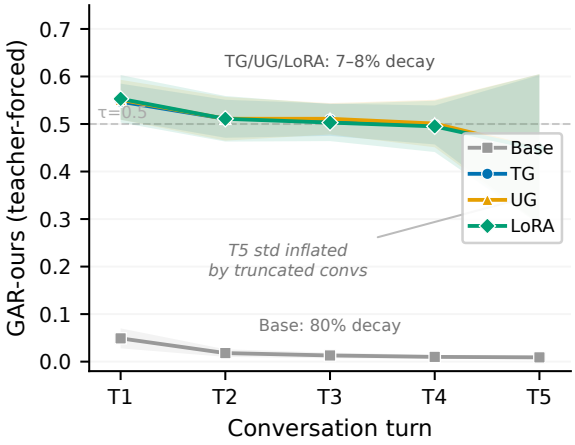


Figure 2: Per-turn GAR-ours under teacher-forced evaluation ($N = 45$, ± 1 std). Fine-tuned variants maintain GAR near $\tau = 0.5$ with 7–8% decay (T1→T4); Base decays 80%. T5 std inflated by truncated conversations. Despite this preservation, autoregressive GAR collapses to 0.015–0.025 for all variants (Table 1), consistent with distribution shift as the primary failure mode under these conditions.

4.3 Inference-Time Attention Steering

All training-time interventions fail uniformly. We test whether inference-time attention manipulation can recover GAR by applying an additive bias β to goal-token attention logits at every layer of the ThreadGate checkpoint during generation.

An initial sweep on $N = 5$ ten-turn conversations (Figure 3a) reveals a sharp nonlinearity between $\beta = 2$ and $\beta = 5$: GAR increases ∼11× ($0.062 \rightarrow 0.6795 \pm 0.020$) while PPL rises 35% ($1.41 \rightarrow 1.91$). At $\beta = 10$, GAR saturates at 0.959 but PPL degrades to 4.22.

$N = 50$ conversations from a held-out evaluation set (39 with ≥3 turns, mean 3.6 user turns; Figure 3b) confirm the qualitative steering pattern observed in the $N = 5$ pilot (10 turns each), with consistent nonlinearity range and saturation behavior. A finer β sweep localizes the nonlinearity: GAR jumps from 0.128 ± 0.110 ($\beta = 2$) to 0.556 ± 0.060

Table 1: Five-way comparison on Qwen-2.5-7B-Instruct. Teacher-forced (TF) evaluation uses IID UltraChat text and 50 held-out conversations; autoregressive (AR) evaluation uses $N = 5$ ten-turn generated conversations. All fine-tuned variants converge under TF (0.75 pp spread); the 20–33× TF-AR gap persists across all five configurations under our evaluation conditions. Base is the pretrained model; all other rows share the same LoRA configuration, optimizer, and training data. GAR-O = GAR-ours; GAR-D = GAR-Dongre (Dongre et al., 2026).

	IID (TF)			Held-out (TF)			AR
Configuration	GAR-O	GAR-D	PPL	GAR-O	GAR-D	PPL	GAR-O
Base	0.01	0.00	4.23	0.01	0.00	4.34	0.012
ThreadGate	0.50	**0.02**	2.78	0.50	**0.02**	2.78	0.016
+PCGrad	0.50	0.01	2.78	0.50	0.01	2.78	0.015
Uniform Gate	**0.51**	0.01	2.78	**0.51**	0.01	2.78	0.015
LoRA-only	0.50	0.01	2.78	0.50	0.01	2.78	**0.025**

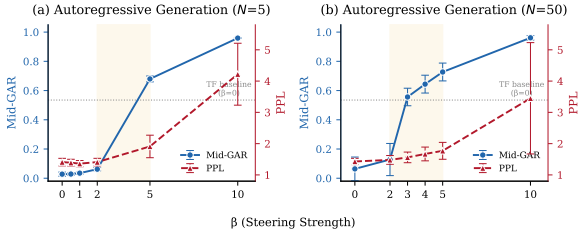


Figure 3: Inference-time attention steering: additive bias β applied to goal-token attention logits during autoregressive generation (error bars show ± 1 std). (a) Initial evaluation ($N = 5$, 10-turn conversations): sharp increase at $\beta \in [2, 5]$, GAR $\sim$11× recovery with 35% PPL rise. (b) Larger-scale evaluation ($N = 50$, 39 with $\geq$3 turns, mean 3.6 user turns) with finer β sweep: the nonlinearity localizes between $\beta = 2$ and $\beta = 3$, identifying $\beta = 3$ as a practical operating point (8.7× GAR recovery, 9% PPL cost). High-β results are consistent across sample sizes. Dotted line: unsteered teacher-forced baseline.

($\beta = 3$), an 8.7× recovery from baseline with only 9% PPL increase (1.43 → 1.56), identifying $\beta = 3$ as a practical operating point. The $\beta = 0$ distribution is right-skewed (mean 0.064, median 0.031, std 0.081), driven by short conversations with high baseline GAR; absolute baselines therefore differ from the 10-turn $N = 5$ evaluation (0.028). At $\beta = 5$, GAR reaches 0.727±0.061 with PPL 1.77±0.27, confirming the recovery pattern at scale.

These results provide constructive evidence that attention misallocation—not representation degradation—is a proximal cause of exposure-bias-induced GAR failure under the tested conditions (Qwen-2.5-7B, UltraChat, GAR-family losses): manually correcting attention via β bias restores GAR, and inference-time intervention circumvents the exposure bias barrier. This steering experiment is diagnostic: it requires identifying goal tokens, trivial for system prompts but not generalizable.

5 Analysis

The five-way comparison in §4 established that gating produces no measurable benefit over LoRA-only fine-tuning. This section traces the failure through a three-step diagnostic chain: we first show that gates learn behavior opposite to the design intent (§5.1), then rule out gradient conflict as the explanation through an exclusion experiment (§5.2), and finally identify distribution shift as the root cause (§5.3). §5.4 confirms this selectivity with an orthogonal hidden-state metric, and §5.5 confirms the failure generalizes across architecturally distinct LLMs.

5.1 Gate Turn Conditioning

The design intent of ThreadGate is to modulate the residual stream more aggressively at late conversation turns, where GAR naturally decays (§3.1). We test whether the learned gates achieve this by comparing gate values at turn 1 (early) and turn 21 (late) across all 28 layers of the trained model.

The gates do the opposite. In 27 of 28 layers, the gate value at turn 1 is lower than at turn 21 (Wilcoxon signed-rank test: $p = 3.73 \times 10^{-8}$, effect size $r = 0.975$). At layer 23, the gate is 0.769 at turn 1 and 0.979 at turn 21; at layer 14, it is 0.747 at turn 1 and 0.928 at turn 21. The pattern is consistent: gates suppress the residual stream more at early turns (where GAR is already high) and become nearly passive at late turns (where intervention is most needed). This inversion has a straightforward explanation. NTP loss pressure is highest at late turns, where the model must predict tokens conditioned on a long context. The NTP gradient pushes gates toward 1.0 to preserve the full sublayer computation, and this pressure is strongest precisely where GAR-based attenuation would be most useful.

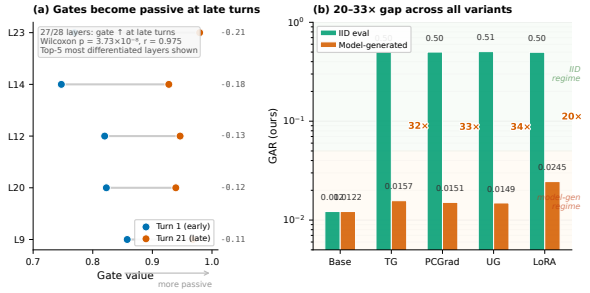


Figure 4: (a) Gate turn conditioning: per-layer gate values at turn 1 vs turn 21. In 27/28 layers, gates approach 1.0 at late turns, inverting the design intent ($p = 3.73 \times 10^{-8}$). (b) Distribution shift: the 20–33× TF-AR GAR gap persists across all five variants (TG, +PCGrad, UG, LoRA-only, Base) under our evaluation conditions.

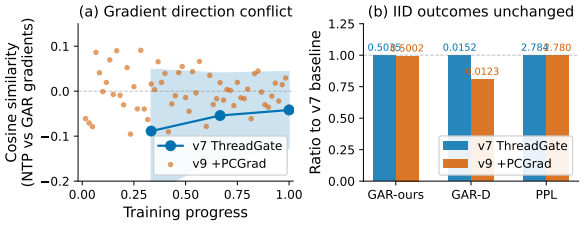


Figure 5: (a) Gradient cosine similarity between NTP and GAR objectives. ThreadGate remains consistently negative; +PCGrad oscillates around zero. (b) IID evaluation outcomes are indistinguishable: eliminating direction conflict changes nothing.

The resulting inference-time gate values range from 0.93 to 0.98 (2–7% attenuation), far below what meaningful modulation would require; this explains the convergence with LoRA-only in Table 1. Figure 4(a) visualizes the per-layer gate difference. The inverted turn conditioning raises a natural question: does the NTP-GAR gradient conflict prevent gates from learning the intended behavior? If gradient conflict is the root cause, resolving it should rescue gate conditioning.

5.2 PCGrad Gradient Conflict Exclusion

The NTP and GAR gradients point in opposing directions across 68–79% of layers throughout training, with mean cosine similarity ranging from -0.089 (epoch 0) to -0.042 (epoch 2). The NTP gradient magnitude dominates by 7.5–14.7×. This raises a natural hypothesis: perhaps opposing gradient directions prevent the GAR objective from converging. If so, eliminating direction conflict should prevent the collapse. We test this using PCGrad (Yu et al., 2020), which projects the gradient of one task onto the normal plane of the other whenever two task gradients conflict. We apply PCGrad to gate parameters only, scoping our claim to direction conflict; magnitude imbalance (Chen et al., 2018) and Nash equilibrium violations (Navon et al., 2022) remain untested.

Eliminating direction conflict via PCGrad shifts per-step cosine similarity from consistently negative (-0.089) to oscillating around zero (-0.15 to $+0.09$), yet all outcomes remain unchanged: IID GAR-ours (0.50 vs 0.50), PPL (2.780 vs 2.784), and autoregressive GAR (0.015 vs 0.016) are indistinguishable (Table 1). $\mathcal{L}_{\text{GAR}}$ follows a nearly identical training trajectory and collapses in the same manner (Figure 5b). The exclusion follows by *modus tollens*: if direction conflict causes the failure, removing it should prevent it; PCGrad removes it; the failure persists; therefore direction conflict is not the cause. Direction conflict is neither necessary (removing it does not rescue GAR) nor sufficient ($\mathcal{L}_{\text{GAR}}$ converges during training despite persistent conflict). The 7.5–14.7× NTP magnitude dominance renders directional interference inconsequential. With gradient conflict ruled out, the question narrows: the training dynamics are not the problem (GAR converges under teacher forcing), so the failure must occur at the transition from training to inference.

5.3 Distribution Shift Analysis

Gates are ineffective (§5.1) and gradient conflict is not the cause (§5.2). The TF-AR GAR gap (Table 1) persists across all training strategies. GAR-Dongre provides a revealing control: under standard training it remains uniformly low (0.01–0.02) with no distribution gap, so the shift selectively destroys the concentrated last-token attention pattern while leaving diffuse attention unaffected (scheduled sampling breaks this pattern; §6).

The mechanism is exposure bias (Bengio et al., 2015; Arora et al., 2022): during training, the model conditions on ground-truth prefixes; at inference, on its own outputs. The statistical properties diverge, and the 2–7% gate attenuation (§5.1) cannot compensate. He et al. (2019) showed that language models exhibit token-level self-recovery from exposure bias, and our PPL results confirm this (Table 1). The critical distinction is between primary and auxiliary objectives: NTP is robust because it is the same task at both training and inference time, while GAR optimizes a distributional property of the input prefix that has no counterpart

under autoregressive generation. Figure 2 confirms teacher-forced GAR-ours remains near $\tau = 0.5$ (7–8% decay, $N = 45$), yet autoregressive GAR collapses for all variants (Figure 4b).

5.4 Mechanism Specificity

§5.3 showed that exposure bias selectively destroys GAR-ours while GAR-Dongre remains unaffected under standard training. We test whether this selectivity extends beyond attention with Turn-level Residual Coherence (TRC), the cosine similarity between hidden states at the last-token position of adjacent assistant turns. For ThreadGate, TF/AR TRC is 0.334/0.332 (gap +0.002); for LoRA-only, 0.349/0.411 (gap −0.062, reversed). Neither shows a gap comparable to the GAR collapse, confirming that exposure bias disrupts a specific routing behavior (summary-position attention to goal tokens) while leaving internal representations intact.

Cross-model evaluation extends this selectivity: GAR-Dongre shows negligible distribution shift on both Llama-3.1-8B (AR/TF ratio 1.002×) and Qwen-2.5-7B (1.28×), compared to the order-of-magnitude collapse in GAR-ours (the scheduled sampling experiment in §6 breaks this pattern, with GAR-Dongre rising to 0.353).

5.5 Cross-Model Generality

To test whether the failure generalizes, we replicate on Llama-3.1-8B-Instruct (Grattafiori et al., 2024) with Base, ThreadGate, and LoRA-only. Phase 1 diagnostics confirm goal attention decay (GAR: 0.026→0.003 over 10 turns). Teacher-forced GAR-ours ($N = 50$) is 0.484 (TG) and 0.481 (LoRA-only); gating contributes zero benefit. We report median GAR-ours for autoregressive evaluation due to heavy-tailed distributions (5–8% of turn-level measurements exhibit sporadic attention spikes). Median GAR-ours ($N = 20$ mid-turn conversations, excluding the first two turns) is 0.018 for both TG and LoRA-only (Base: 0.009), yielding TF-AR gaps of 27× and 26×. The TF-AR gap exceeds 25× on both architectures, consistent with the exposure bias hypothesis. The gate profile differs—Qwen's minimum at layers 3–4, Llama's at layer 25—yet both converge toward identity under NTP pressure.

6 Discussion

The failure traces to an incompatibility between teacher-forcing and GAR-family auxiliary losses, not to the gating architecture. NTP gradient pressure (7.5–14.7× the GAR magnitude) pushes gates toward identity (2–7% attenuation), while inference-time steering ($\beta = 5$) recovers GAR from 0.064 to 0.727, providing constructive evidence that attention misallocation is a proximal cause (§4.3). Alternative hypotheses remain open; our evidence is consistent with exposure bias as the dominant factor but does not exclude alternatives. In particular, KV cache drift—accumulation of model-generated key-value states diverging from teacher-forced statistics—could compound the mismatch; disentangling prefix-level and cache-level contributions is left to future work. Split-softmax (Li et al., 2024) and Rhea (Hong et al., 2025) complement β-steering by restructuring attention computation at the architectural level, but cannot resolve exposure bias in auxiliary losses requiring retraining.

Scheduled sampling as training-time mitigation. A preliminary experiment (single seed) applies scheduled sampling (Bengio et al., 2015), mixing teacher-forced and model-generated contexts. AR GAR-ours increases from 0.016 to 0.318 (20×), compressing the ∼31× gap to 1.57× with negligible PPL change (1.38 vs. 1.36), consistent with exposure bias as a primary driver. GAR-Dongre also rises (0.019→0.353). On the $N = 50$ evaluation, $\beta = 5$ incurs 24% PPL cost (1.43 → 1.77); $\beta = 3$ offers a more balanced trade-off (+9%); scheduled sampling incurs no fluency cost but requires retraining. A second seed yields AR GAR = 0.418 (gap 1.20×); systematic ablation is deferred. Exploratory instruction retention analysis (Appendix C) shows +23.8 pp compliance at $\beta = 3$ (McNemar $p = 0.077$).

TRC analysis (§5.4) confirms the mismatch targets attention routing, not representations. We scope claims to GAR losses on Qwen-2.5-7B and Llama-3.1-8B; magnitude imbalance (Chen et al., 2018) and Nash equilibria (Navon et al., 2022) remain untested.

7 Conclusion

ThreadGate modulates the residual stream with per-layer gated MLP networks and an auxiliary GAR loss. Training succeeds (TF GAR = 0.50), but autoregressive GAR collapses by >25× on both Qwen-2.5-7B and Llama-3.1-8B. Steering ($\beta = 5$, GAR 0.727) and scheduled sampling (gap 31×→1.57×; single seed) implicate attention misallocation as the proximal cause.

8 Limitations

The five-way comparison uses seed = 42; ThreadGate is additionally validated with seed = 123 (TF/AR GAR-ours within ± 0.001, gap 30.5–32.1×). The remaining three variants use a single seed; cross-variant convergence provides indirect robustness evidence but does not substitute for multi-seed validation.

We test only one type of auxiliary attention loss (GAR with MSE target $\tau = 0.5$). Other attention steering objectives, such as head-specific or layer-specific attention losses, may exhibit different sensitivity to exposure bias. Our negative result applies to the GAR family; it does not rule out all auxiliary attention interventions.

Training uses 8K conversations from UltraChat. Larger-scale training could alter gate dynamics, particularly if the model encounters sufficient distributional diversity to reduce the training-inference mismatch.

All experiments use two model families (Qwen-2.5-7B and Llama-3.1-8B); while cross-model evaluation confirms the exposure bias gap, $N = 2$ does not establish universality. The Llama evaluation includes ThreadGate and LoRA-only; the full five-way ablation (Uniform Gate, +PCGrad) has not been replicated, and all experiments use a single dataset (UltraChat). The instruction retention analysis is exploratory: a single model, 18 system-prompt constraint types, and a single β value without grid search.

The PCGrad exclusion experiment addresses gradient direction conflict only. Magnitude imbalance (GradNorm; Chen et al., 2018) and Nash equilibrium approaches (Nash-MTL; Navon et al., 2022) remain untested and could reveal additional failure modes or interaction effects.

All four fine-tuned variants share the same GAR auxiliary loss. This design isolates the gating mechanism but does not test whether gating could be effective without an auxiliary attention loss or under a different training objective.

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A Training Hyperparameters

All fine-tuned variants share the following configuration: Qwen-2.5-7B-Instruct base model, LoRA rank 16 (α=32) on `q_proj` and `v_proj`, AdamW optimizer (lr = 2×10^{-4}, β_1=0.9, β_2=0.999), batch size 1 with gradient accumulation 8, 3 epochs over 8K UltraChat conversations, maximum sequence length 2048, seed 42. ThreadGate gates use a $10\times$ learning rate multiplier and initial bias 1.5 ($\sigma(1.5) \approx 0.82$).

B Loss Formulation Search

Table 2 summarizes the seven loss configurations explored before arriving at the final MSE + gate diversity formulation (§3.3). Configurations v1–v6 span three alternative loss families; none converged during training.

C Instruction Retention Details

Across 18 system-prompt constraint types, $\beta = 3$ steering increases content constraint compliance by +23.8 pp (GAR-compliance $r = 0.200$, $p = 0.007$; McNemar $p = 0.077$), providing exploratory evidence for downstream benefit pending multi-model confirmation.

Maximize-family losses ($-\text{mean}(\text{GAR})$) consistently drove gates toward identity: the unbounded objective rewards full pass-through ($g \to 1.0$), collapsing gate diversity (std < 0.025). Bounded variants (smooth saturation) decelerated but did not prevent this collapse, motivating the MSE formulation targeting a finite GAR level.

D Additional Visualizations

D.1 Gate Profile Heatmap

D.2 Cross-Model Gate Profile Comparison

D.3 Per-Turn AR GAR Decay

D.4 Gate Value Distributions

D.5 Training Loss Curves

D.6 Full Attention Steering Results

Table 2: Loss formulation iterations on Qwen-2.5-7B-Instruct. GAR-ours (teacher-forced) and PPL are reported only for the converged configuration (v7).

	Loss Family	Status	Failure Mechanism	GAR	PPL
v1	Cumulative	× GAR loss → 0	GAR loss collapses to zero once threshold is reached; gate gradients vanish at equilibrium	—	—
v2	Cumulative + ε	× same trajectory	Numerical instability from cumulative product; ε-stabilization insufficient	—	—
v3	Maximize (λ=2)	× gate frozen	Unbounded reward drives gates toward identity (all-pass); no layer selectivity	—	—
v3b	Maximize (λ=10)	× identity collapse	Higher λ accelerates convergence to same degenerate identity solution	—	—
v5	Maximize (λ=1)	× identity collapse	Rebalancing delays but does not prevent identity collapse	—	—
v6	Smooth saturation	× identity collapse	Monotonic reward identical to maximize in practice; boundedness slows but does not alter terminal trajectory	—	—
v7	MSE + gate div	✓ **converged**	**Bidirectional penalty prevents identity collapse; gate diversity regularizer enforces layer selectivity**	0.50	2.78

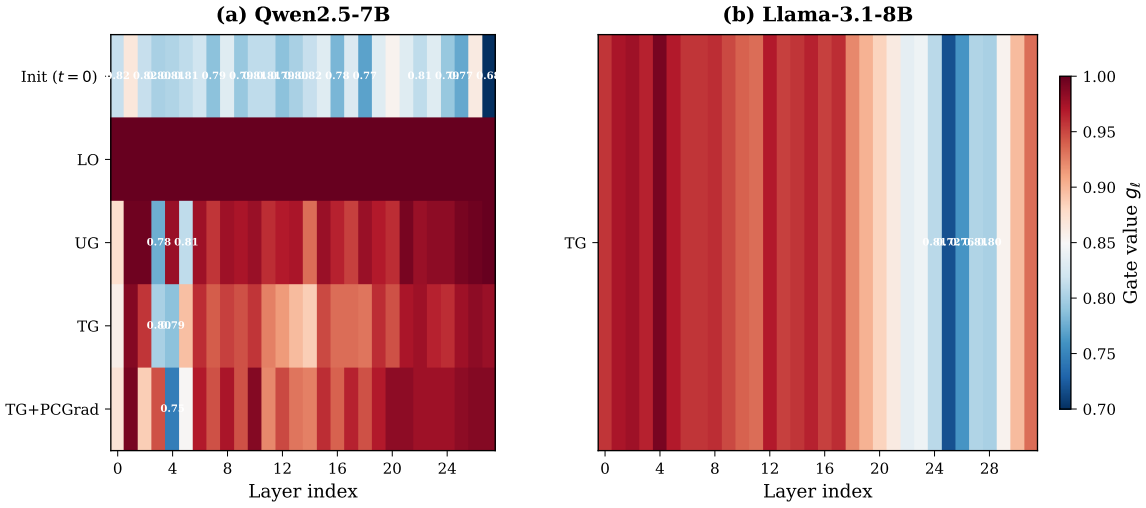


Figure 6: Per-layer gate values across training for Qwen-2.5-7B (28 layers × 5 conditions) and Llama-3.1-8B (32 layers × 1 condition). Early layers (L1, L4–5) consistently receive the lowest gate values across conditions.

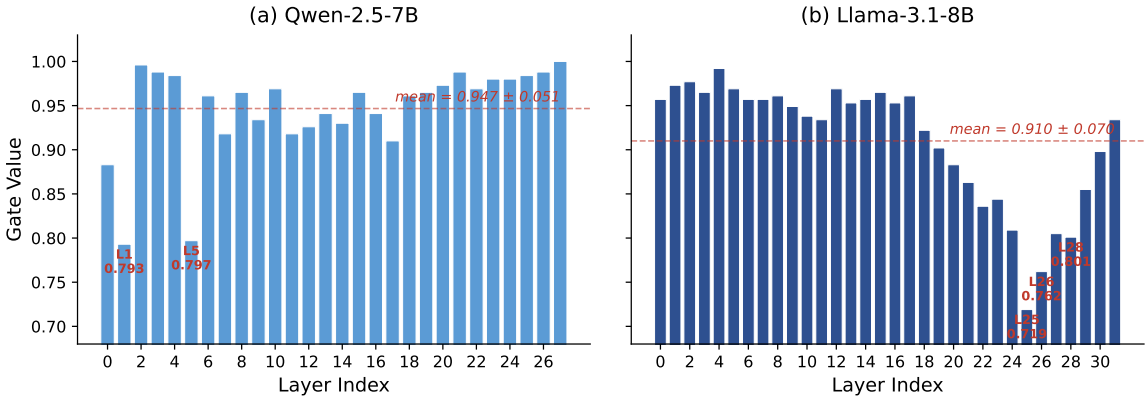


Figure 7: Side-by-side gate profiles for Qwen-2.5-7B and Llama-3.1-8B. Qwen dips at early layers (L1 = 0.793, L5 = 0.797), while Llama shows a V-shaped valley at mid-to-late layers (L25 = 0.719). Both models learn layer-selective gating but with architecture-dependent patterns.

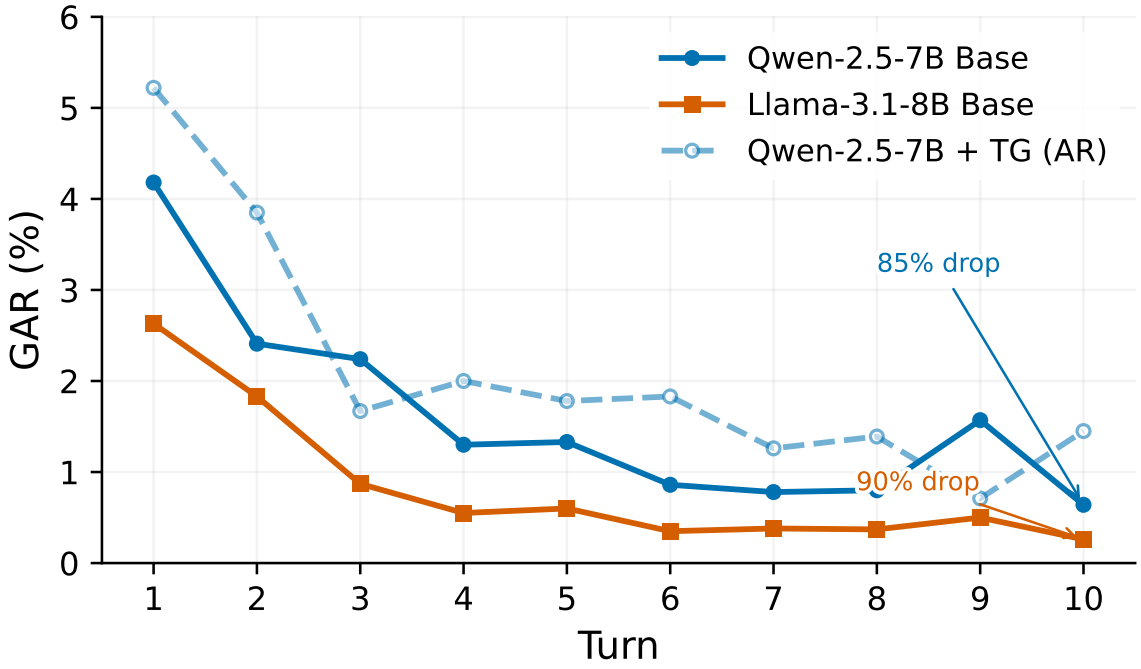


Figure 8: GAR decays monotonically across turns under autoregressive generation. Qwen base drops 85% (turn 1→10), Llama base drops 90%. The ThreadGate-trained model (AR) shows slight elevation but the same decay pattern, confirming that training-time GAR optimization does not transfer to generation.

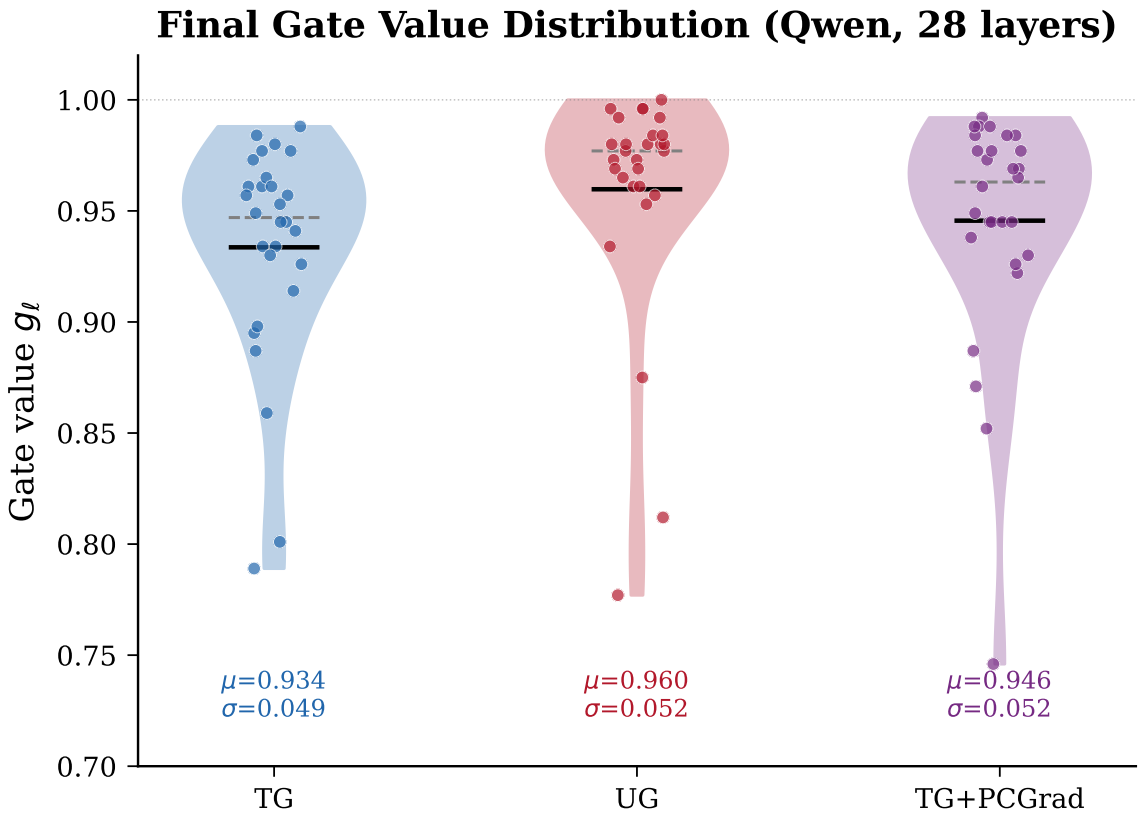


Figure 9: Per-layer gate value distributions for ThreadGate, Uniform Gate, and PCGrad variants. All three converge to similar distributions (mean ∼0.94–0.96), confirming zero incremental benefit of the gating mechanism over LoRA-only.

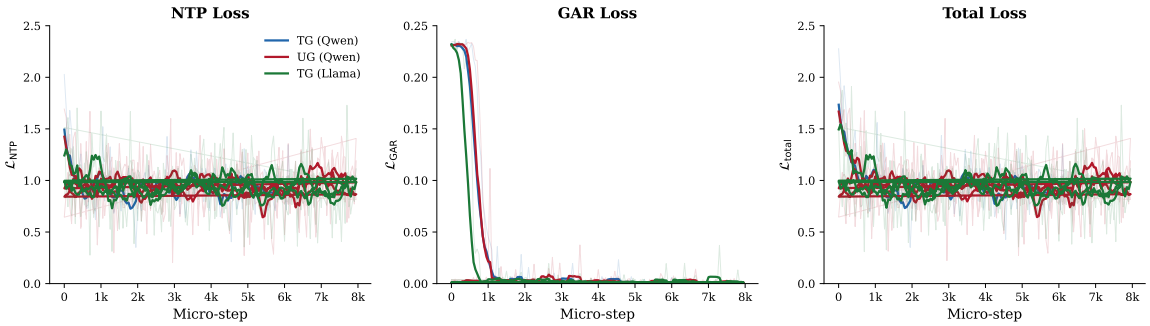


Figure 10: NTP, GAR, and total loss trajectories. GAR loss collapses to near-zero within the first epoch across all variants, while NTP loss converges normally.

Table 3: Full attention steering results across evaluation settings. **Bold** indicates the operating point cited in the main text ($\beta = 5$). A visual gap separates the phase-transition boundary: GAR jumps sharply between $\beta = 2$ and $\beta = 3$ across all settings.

	$N = 5$, Gen		$N = 50$, Gen		$N = 200$, TF	
β	GAR	PPL	GAR	PPL	GAR	PPL
0	.028	1.41	$.064 \pm .081$	$1.43 \pm .10$	.535	2.62
0.5	.029	1.39	—	—	—	—
1	.035	1.36	—	—	—	—
2	.062	1.41	$.128 \pm .110$	$1.48 \pm .14$	.611	2.69
3	—	—	$.556 \pm .060$	$1.56 \pm .17$	.788	2.93
4	—	—	$.644 \pm .061$	$1.67 \pm .22$	.817	3.49
5	**.680**	**1.91**	$\mathbf{.727 \pm .061}$	$\mathbf{1.77 \pm .27}$	**.833**	**6.15**
10	.959	4.22	$.961 \pm .015$	3.45 ± 1.78	.977	505.2